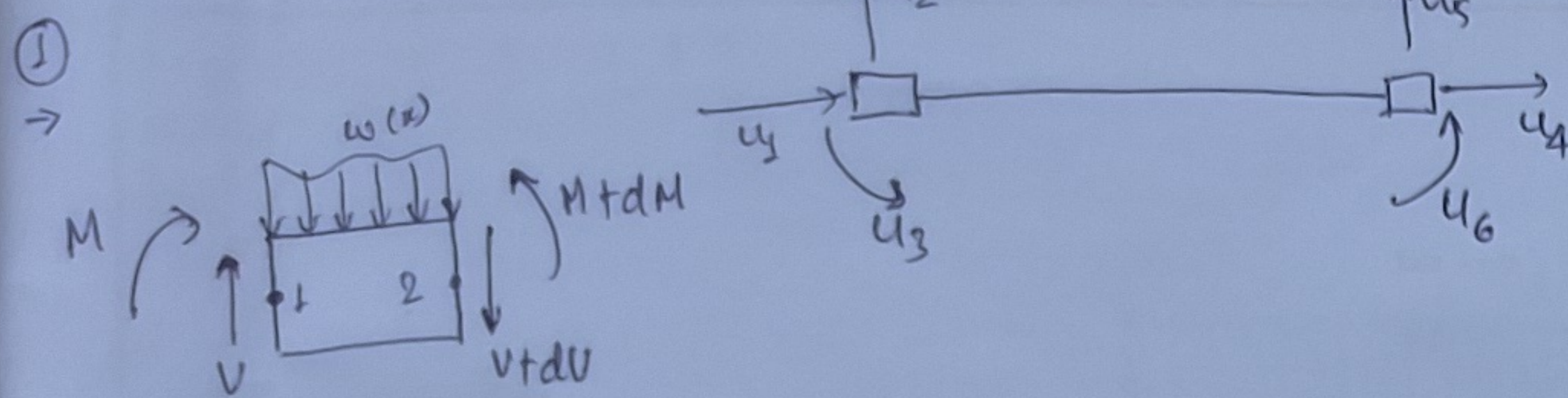


# Homework 1



## Equilibrium:

$$\sum F_y = 0 \Rightarrow V - (V+dv) - w(x)dx = 0$$

$$\Rightarrow \frac{dV}{dx} = -w(x) \quad \text{--- (i)}$$

$$\sum M_2 = 0 \Rightarrow Vdx + M - (M+dm) - w(x) \cdot dx \left(\frac{dx}{2}\right) = 0$$

$$\Rightarrow \frac{dM}{dx} = V \quad \text{--- (ii)}$$

## Constitutive:

$$\text{Curvature } (K) = \frac{1}{\rho} = \frac{M}{EI} \quad \text{--- (iii)}$$

## Compatibility:

for small slope,

$$\tan \phi \approx \phi \approx \frac{dy}{dx}$$

$$\Rightarrow \phi = \frac{dy}{dx} \quad \text{--- (iv)}$$

$$K \approx \frac{d^2y}{dx^2} \approx \frac{d\phi}{dx} \quad (\text{rate of change of slope}) = \frac{d^2y/dx^2}{\left(1 + \left(\frac{dy}{dx}\right)^2\right)^{3/2}}$$

from (iii) & (iv),

$$\Rightarrow \frac{d^2y}{dx^2} = \frac{M}{EI} \quad \text{--- (v)}$$

$$\Rightarrow M = EI \frac{d^2y}{dx^2} \quad \text{--- (va)}$$

$$\Rightarrow \frac{dM}{dx^2} = \frac{d^2}{dx^2} \left( EI \frac{d^2y}{dx^2} \right) = -w(x)$$

And,

$$\frac{dM}{dx} = V = \frac{d}{dx} EI \frac{d^2y}{dx^2} \quad \text{--- (vi)}$$

for constant EI,  $V = EI \frac{d^3y}{dx^3}$  and if only nodal

$$\text{forces and moments exist, } w(x) = 0 \Rightarrow \frac{d^4y}{dx^4} = 0 \quad \text{--- (vii)}$$

## Displacement Function

assume,  $y(x) = a_1x^3 + a_2x^2 + a_3x + a_4 = [x^3 \ x^2 \ x \ 1] \begin{Bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{Bmatrix}$

Cubic function for displacement,

$\rightarrow$  4 DOFs

$\rightarrow$  satisfies basic beam eqn (vii)

$$\Rightarrow \phi(x) = 3a_1x^2 + 2a_2x + a_3$$

Boundary Conditions:

At  $x=0$ ,

$$y(0) = u_2 = a_4$$

$$\frac{dy(0)}{dx} = u_3 = a_3$$

At  $x=L$ ,

$$y(L) = u_5 = a_1L^3 + a_2L^2 + a_3L + a_4$$

$$\frac{dy(L)}{dx} = u_6 = 3a_1L^2 + 2a_2L + a_3$$

$$\begin{Bmatrix} u_2 \\ u_3 \\ u_5 \\ u_6 \end{Bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ L^3 & L^2 & L & 1 \\ 3L^2 & 2L & 1 & 0 \end{bmatrix} \begin{Bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{Bmatrix} = \{d\}$$

$$y = [N] \{d\} = N_1u_2 + N_2u_3 + N_3u_5 + N_4u_6$$

$[N] = [N_1 \ N_2 \ N_3 \ N_4]$  = shape functions or interpolation functions known as Hermite cubic interpolation functions.

$$N_1 = \frac{1}{L^3} (2x^3 - 3x^2L + L^3) \quad N_2 = \frac{1}{L^3} (x^3L - 2x^2L^2 + xL^3)$$

$$N_3 = \frac{1}{L^3} (-2x^3 + 3x^2L) \quad N_4 = \frac{1}{L^3} (x^3L - x^2L^2)$$

Using (va), (vi),

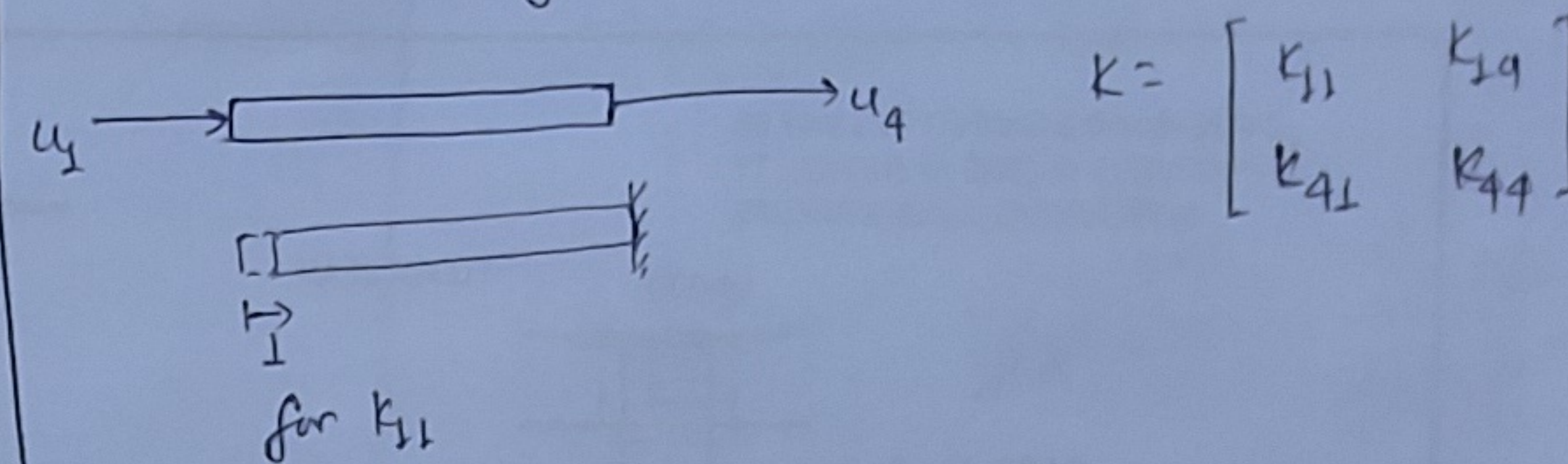
$$f_{1y} = V = EI \frac{d^3y(0)}{dx^3} = \frac{EI}{L^3} (12u_2 + 6Lu_3 - 12u_5 + 6Lu_6)$$

$$m_1 = -m = -EI \frac{d^2y(0)}{dx^2} = \frac{EI}{L^3} (6Lu_2 + 4L^2u_3 - 6Lu_5 + 2L^2u_6)$$

$$f_{2y} = -V = -EI \frac{d^3y(L)}{dx^3} = \frac{EI}{L^3} (-12u_2 - 6Lu_3 + 12u_5 - 6Lu_6)$$

$$m_2 = m = EI \frac{d^2y(L)}{dx^2} = \frac{EI}{L^3} (6Lu_2 + 2L^2u_3 - 6Lu_5 + 4L^2u_6)$$

Consider axial deformation,



$$\Delta = \frac{PL}{AE}$$

for  $\Delta=1, P=k$

$$\Rightarrow k = \frac{AE}{L}$$

$$\Rightarrow K = \begin{bmatrix} k & -k \\ -k & k \end{bmatrix}$$

$$\begin{Bmatrix} f_{1x} \\ f_{2x} \end{Bmatrix} = \frac{AE}{L} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \end{Bmatrix}$$

for  $K_{11}$ , fix all coordinates and apply unit deflection at 1,

$$\Rightarrow K_{11} = k, \quad K_{12} = -k$$

$\rightarrow$  cause ( $\Delta$ )  
 $\rightarrow$  effect (P)

Combining the ~~stiffness~~ matrices,

As  $F = KQ$

$$\begin{Bmatrix} f_{1x} \\ f_{1y} \\ m_1 \\ f_{2x} \\ f_{2y} \\ m_2 \end{Bmatrix} = \begin{bmatrix} AE/L & 0 & 0 & -AE/L & 0 & 0 \\ 0 & 12EI/L^3 & 6EI/L^2 & 0 & -12EI/L^3 & 6EI/L^2 \\ 0 & 6EI/L^2 & 4EI/L & 0 & -6EI/L^2 & 2EI/L \\ -AE/L & 0 & 0 & AE/L & 0 & 0 \\ 0 & -12EI/L^3 & -6EI/L^2 & 0 & 12EI/L^3 & -6EI/L^2 \\ 0 & 6EI/L^2 & 2EI/L & 0 & -6EI/L^2 & 4EI/L \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \\ u_6 \end{Bmatrix}$$

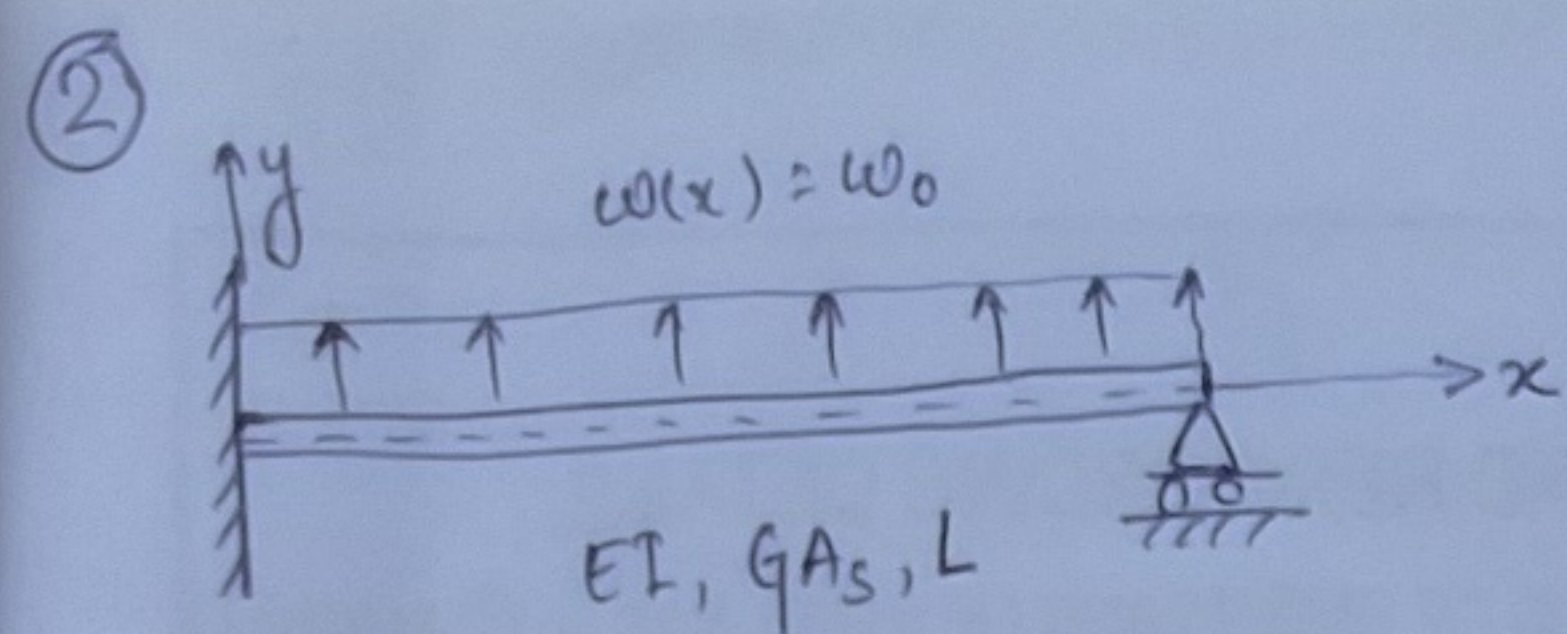
Stiffness Matrix is symmetric because,

$\rightarrow$  work done by force A moving through displacement B equals the work done by force B moving through displacement A.

$\rightarrow$  Proved by Betti's Reciprocal Theorem

$$F_1 \cdot u_{12} = F_2 \cdot u_{21}, \quad K_{ij} = K_{ji}$$

$\rightarrow$  In inelastic systems, the stiffness matrix is generally NOT symmetric because the energy argument fails.



Sol<sup>n</sup>:

Considering Timoshenko Beam Theory,

① Total deflection at a point,

$$\frac{dy}{dx} = \alpha(x) + \gamma(x) \Rightarrow \gamma(x) = -\frac{dy}{dx} + \alpha(x) \quad \text{--- (I)}$$

$\downarrow$  due to shear force  
 $\downarrow$  due to bending

Assume linear deflection, angular deflection (slope) are small,

① Relation between bending moment and bending deformation (curvature)

$$M(x) = EI \frac{d\alpha(x)}{dx} \quad \text{--- (II)}$$

② Relation between shear force and shear deformation (Shear strain)

$$V(x) = A_s G \gamma(x) \quad \text{--- (III)}$$

③ Equilibrium,

$$\frac{dM}{dx} = V \quad \text{--- (IV)}, \quad \frac{dV}{dx} = -w \quad \text{--- (V)}$$

from (I), (IV), (V),

$$EI \frac{d^3\alpha}{dx^3} = w(x) \quad \text{--- (VI)}$$

from (I), (III),

$$\frac{dy}{dx} = \alpha(x) - \frac{V(x)}{A_s G} = \alpha(x) - \frac{EI}{GA_s} \frac{d^2\alpha}{dx^2} \quad \text{--- (VII)}$$

④ Shape Function,

to satisfy eq<sup>n</sup> (VI),

$$\alpha(x) = C_0 + C_1 x + C_2 x^2 + \frac{w_0}{6EI} x^3 \quad \text{--- (VIII)}$$

$$\begin{aligned} \frac{dy}{dx} &= C_0 + C_1 x + C_2 x^2 + \frac{w_0}{6EI} x^3 - \frac{EI}{GA_s} \left( 2C_2 + \frac{w_0}{EI} x \right) \\ &= \left( C_0 - \frac{2EI C_2}{GA_s} \right) + \left( C_1 - \frac{w_0}{GA_s} \right) x + C_2 x^2 + \frac{w_0}{6EI} x^3 \end{aligned}$$

$$\Rightarrow y(x) = \left( C_0 - \frac{2EI C_2}{GA_s} \right) x + \left( C_1 - \frac{w_0}{GA_s} \right) \frac{x^2}{2} + C_2 \frac{x^3}{3} + \frac{w_0}{6EI} \left( \frac{x^4}{4} \right) + C_3 \quad \text{--- (IX)}$$

⑤ Boundary Conditions,

At  $x=0, y=0, \alpha=0$

$$y=0 \Rightarrow C_3=0 \quad \text{--- (a)}$$

$$\alpha=0 \Rightarrow C_0=0 \quad \text{--- (b)}$$

At  $x=L, y=0, M=0$

$$y=0 \Rightarrow 0 = \left( -\frac{2EI C_2}{GA_s} \right) L + \left( C_1 - \frac{w_0}{GA_s} \right) \frac{L^2}{2} + C_2 \frac{L^3}{3} + \frac{w_0}{6EI} \frac{L^4}{4}$$

$$M=0 \Rightarrow EI \frac{d\alpha(L)}{dx} = 0 = C_1 + 2C_2 L + \frac{w_0}{6EI} 3L^2$$

$$\Rightarrow C_1 = -2C_2 L - \frac{w_0 L^2}{2EI}$$

$$\Rightarrow \left( -\frac{2EI L}{GA_s} \right) C_2 + \left( -2C_2 L - \frac{w_0 L^2}{2EI} \right) \frac{L^2}{2} - \frac{w_0 L^2}{2GA_s} + C_2 \frac{L^3}{3} + \frac{w_0 L^4}{24EI} = 0$$

$$\Rightarrow C_2 \left( -\frac{2EIL}{GA_s} - L^3 + \frac{L^3}{3} \right) - \frac{w_0 L^4}{4EI} - \frac{w_0 L^2}{2GA_s} + \frac{w_0 L^4}{24EI} = 0$$

$$C_2 = \left( \frac{w_0 L^2}{2GA_s} + \frac{5}{24} \cdot \frac{w_0 L^4}{EI} \right) / \left( -\frac{2EIL}{GA_s} - \frac{2L^3}{3} \right) \quad \text{--- (c)}$$

$$\Rightarrow C_1 = -2L \left( \frac{\frac{w_0 L^2}{2GA_s} + \frac{5}{24} \cdot \frac{w_0 L^4}{EI}}{-\frac{2EIL}{GA_s} - \frac{2L^3}{3}} \right) + \frac{w_0 L^2}{2EI} \quad \text{--- (d)}$$

Thus,

$$y(x) = -\frac{2EI}{GA_s} C_2 x + C_1 \frac{x^2}{2} - \frac{w_0}{GA_s} \frac{x^2}{2} + C_2 \frac{x^3}{3} + \frac{w_0}{6EI} \frac{x^4}{4} \quad \text{--- (I)}$$

$$\alpha(x) = C_1 x + C_2 x^2 + \frac{w_0}{6EI} x^3 \quad \text{--- (2)}$$

$$\gamma(x) = \frac{2EI}{GA_s} C_2 + \frac{w_0}{GA_s} x \quad \text{--- (3)}$$

Unit check,

$$C_2 = \left( \frac{N/m \cdot x \cdot m^2}{N/m^2 \cdot x \cdot m^2} - \frac{N/m \cdot m^4}{N/m^2 \cdot x \cdot m^4} \right) / \left( -\frac{N/m^2 \cdot m^4}{N/m^2 \cdot m^2} \right)$$

$$m/m^3 = 1/m^2$$

$$C_1 = -\frac{1}{m} x \cdot m^2 + \frac{N/m \cdot m^4}{N/m^2 \cdot x \cdot m^4} \cdot \frac{N}{m^2} \cdot x \cdot m^2$$

$$= 1/m$$

$$\alpha = \frac{1}{m} x \cdot m + \frac{1}{m^2} x \cdot m^2 + \frac{N/m}{N/m^2 \cdot x \cdot m^2} x \cdot m^3$$

$$y = m^2 x \cdot \frac{1}{m^2} x \cdot m + \frac{N/m}{N/m^2 \cdot x \cdot m^2} x \cdot m^3$$

$$M(x) = EI (C_1) + (2C_2) x EI + \frac{w_0}{2} x^2$$

$$V(x) = 2EI C_2 + w_0 x \quad \text{--- (5)}$$

for  $G A_s \rightarrow \infty$  i.e. Euler-Bernoulli Beam,

$$C_2 = \frac{+\frac{5}{24} \frac{w_0 L^4}{EI}}{-\frac{2L^3}{3}} \quad C_1 = -2L \times \frac{5}{16} \frac{w_0 L}{EI} = -\frac{w_0 L^2}{2EI}$$

$$= -\frac{5}{16} \frac{w_0 L}{EI} \quad \Rightarrow C_1 = \frac{1}{8} \frac{w_0 L^2}{EI}$$

$$C_1 = -\frac{5}{16} \frac{w_0 L}{EI}$$

$$y(x) = \frac{-2EI}{G A_s} \left( -\frac{5}{16} \frac{w_0 L}{EI} \right) x + \left( \frac{1}{8} \frac{w_0 L^2}{EI} \right) \frac{x^2}{2} - \frac{w_0}{G A_s} \left( \frac{x^2}{2} \right)$$

$$+ \left( \frac{5}{16} \frac{w_0 L}{EI} \right) \frac{x^3}{3} + \frac{w_0}{6EI} \left( \frac{x^4}{4} \right)$$

$$\Rightarrow y(x) = +\frac{1}{16EI} w_0 L^2 x^2 - \frac{5}{48EI} w_0 L x^3 + \frac{w_0}{24EI} x^4 \quad (1a)$$

$$\alpha(x) = +\frac{1}{8EI} w_0 L^2 x - \frac{5}{16EI} w_0 L x^2 + \frac{w_0}{6EI} x^3 \quad (2a)$$

$$\gamma(x) = 0 \quad (3a)$$

$$M(x) = -\frac{w_0 L^2}{8} + \frac{5}{8} w_0 L x + \frac{w_0}{2} x^2 \quad (4a)$$

$$V(x) = -\frac{5}{8} w_0 L + w_0 x \quad (5a)$$

for  $EI \rightarrow \infty$  i.e. pure shear beam,

$$\frac{2EI}{G A_s} \times \frac{\frac{w_0 L^2}{24} + \frac{5}{24} \frac{w_0 L^4}{EI}}{-\frac{2L^3}{3}} = \frac{2}{G A_s} \times \frac{\frac{w_0 L^2}{2}}{-\frac{2L^3}{3}} = -\frac{w_0 L}{24 G A_s}$$

$$= \frac{2EI}{G A_s} C_2$$

$$C_1 = 0, C_2 = 0$$

Thus,

$$y(x) = +\frac{w_0 L}{24 G A_s} x - \frac{w_0 x^2}{24 G A_s} \quad (1b)$$

$$\alpha(x) = 0 \quad (2b)$$

$$\gamma(x) = -\frac{w_0 L}{24 G A_s} + \frac{w_0}{24 G A_s} x \quad (3b)$$

$$M(x) = -\frac{w_0 L}{2} x + \frac{w_0 x^2}{2} \quad (4b)$$

$$V(x) = -\frac{w_0 L}{2} + w_0 x \quad (5b)$$

$$EI C_1 = -2L(EI C_2) + \frac{w_0 L^2}{2}$$

$$= -2L \left( -\frac{w_0 L}{24 G A_s} \right) - \frac{w_0 L^2}{2}$$

$$= 0$$

$$2EI C_2 = -\frac{w_0 L}{2}$$

3

$$E = 29000 \text{ ksi}$$

$$\nu = 0.3$$

$$G = \frac{E}{2(1+\nu)}$$

Beam section (W18X175):  $h = 20.04 \text{ in}$ ,  $I = 3450 \text{ in}^4$ ,  $A_s = 17.8 \text{ in}^2$

load( $w_0$ ) = 2 kips/in

$$L = [2, 5, 10] \text{ ft}$$

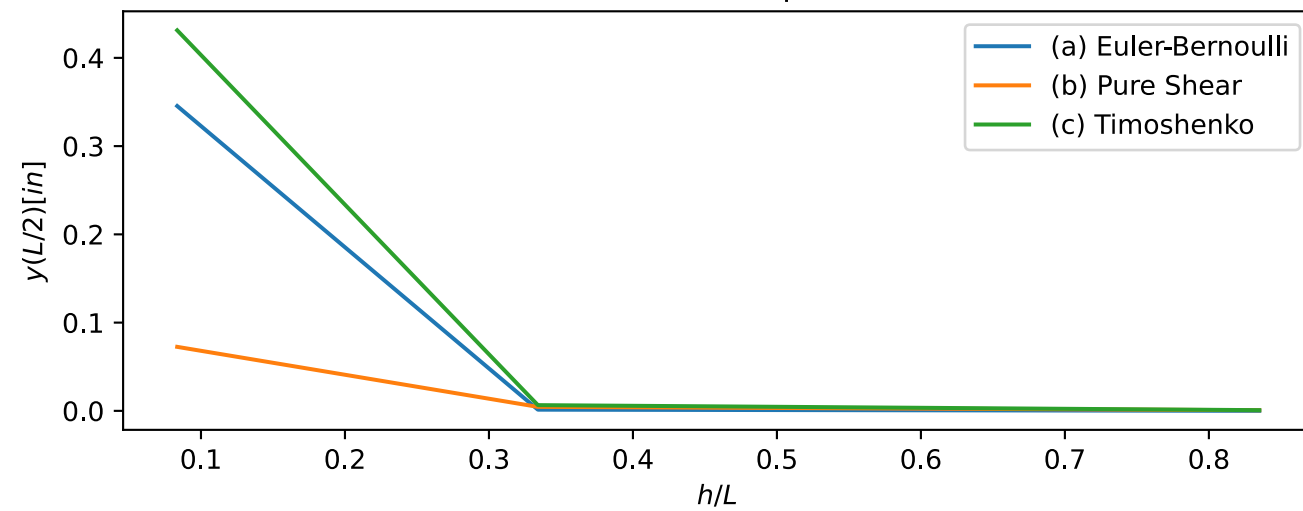
To plot  $y(x)$ ,  $V(x)$ ,  $M(x)$  as a function of  $\frac{h}{L}$  and  $\phi = \frac{12EI}{G A_s L^2}$

for three cases,

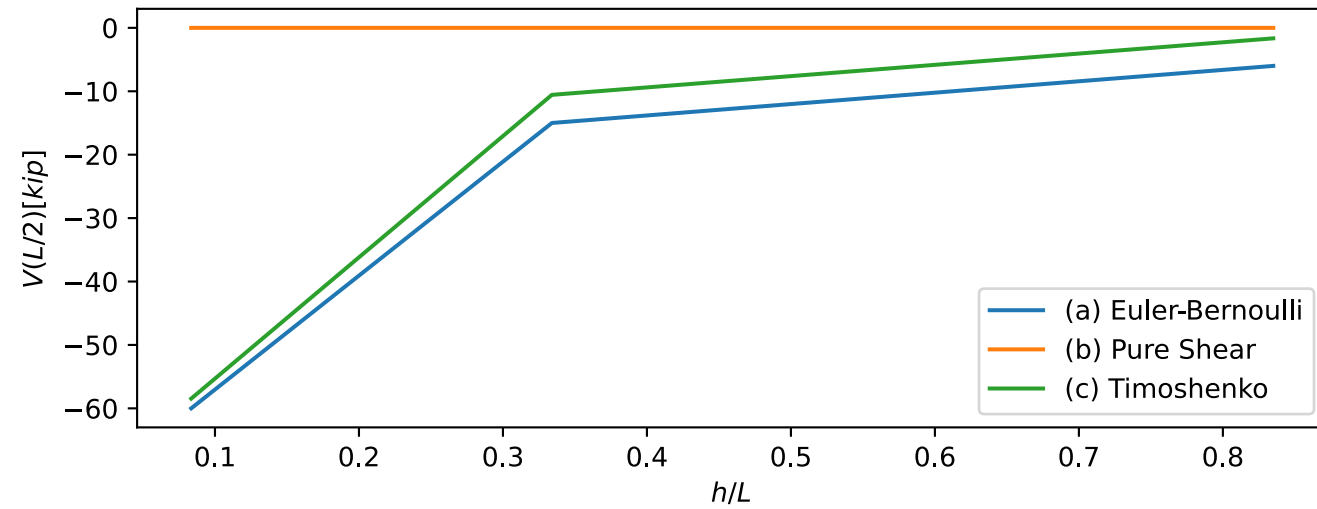
### Questions

- Quantify Eq (1) to (5) in terms of  $\phi$ ,  $h/L$
- Interpret the figures/plots

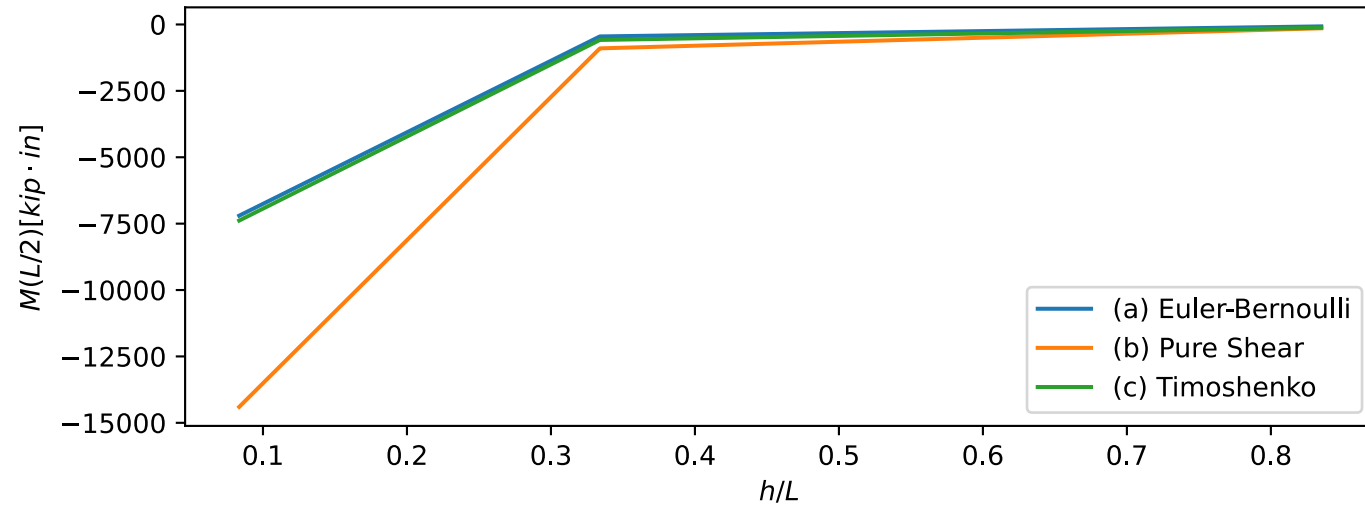
Deflection Comparison



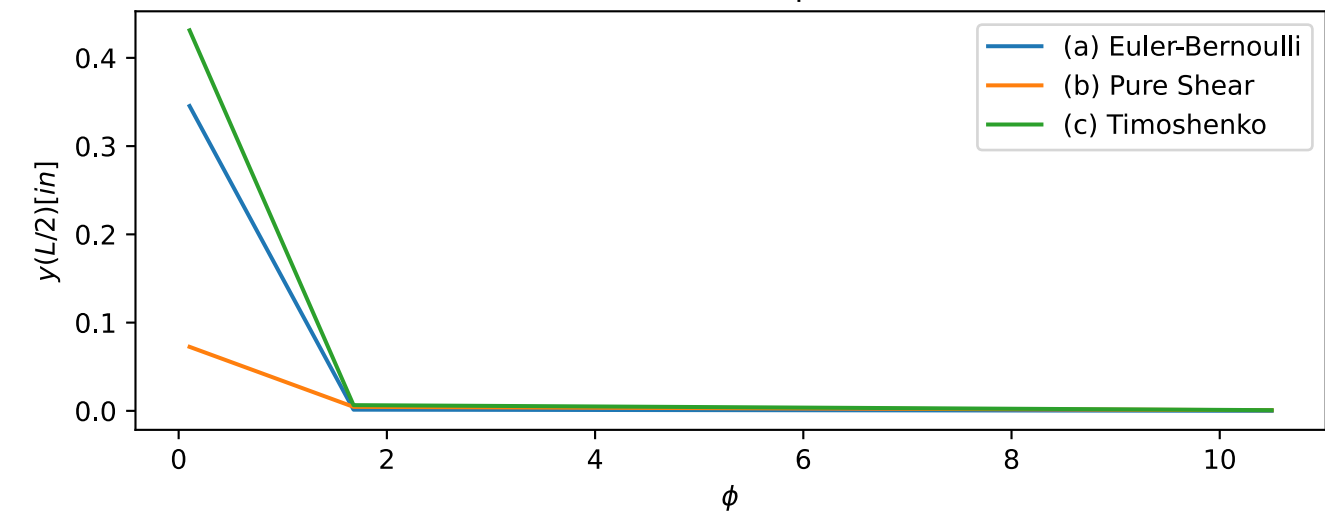
Shear Force Comparison



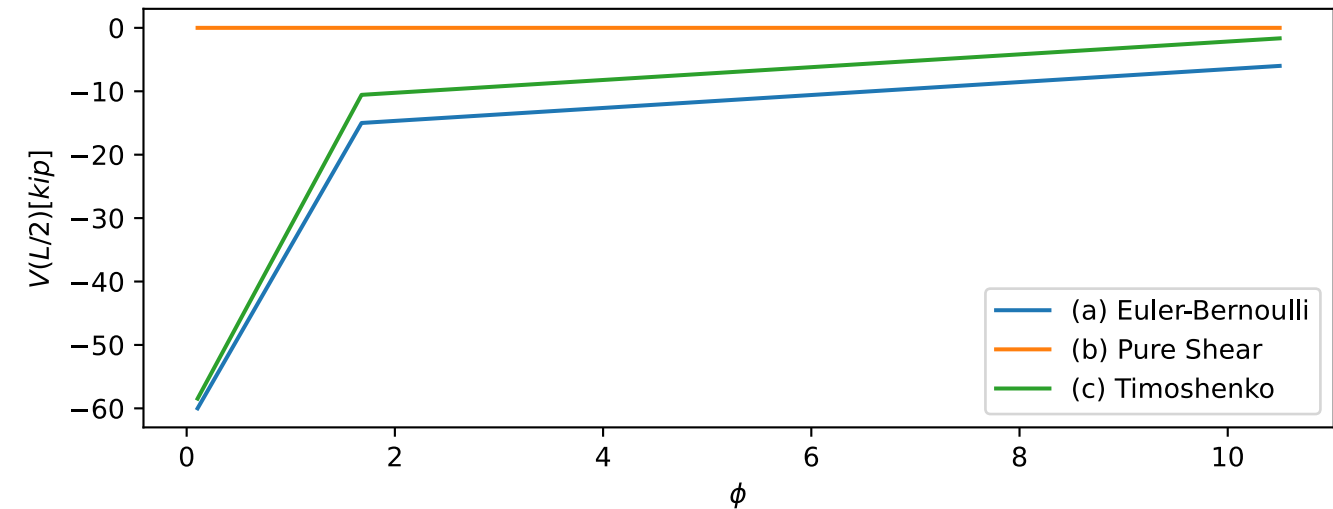
Bending Moment Comparison



Deflection Comparison



Shear Force Comparison



Bending Moment Comparison

